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AI DETECTION REPORT — 2026-05-29

Very likely human-written

0%

LOW LIKELIHOOD OF AI

0%

AI-generated

0%

AI-assisted / paraphrased

100%

Likely human

Analysis

- Average sentence length in human writing is closer to 23.2 words, which matches the sample’s generally moderate sentence spans rather than AI’s 29.2; the excerpt shows a mix of moderate-length sentences, consistent with human style in the 23-word vicinity (Research comparison: AI 29.2 vs Human 23.2).
— *“Waste management involves managing a variety of waste materials like gases, liquids or solids.”*
- There are clearly multiple very short sentences and bullet-like statements scattered through the excerpt, consistent with human writing’s tendency to include concise lines in procedural sections (Human-signal pattern: 5.8% very short sentences vs AI 2.4%).
— *“Images of different waste belonging to the following classes are available in”*
- High function word usage pattern: The text shows frequent function words in connective phrases and descriptions, consistent with the human-skewed function word ratio around 40%, in contrast to AI’s 33%.
— *“The goal of this project is to develop an intelligent system which is capabl”*
- Longer mean word length with a mix of common terms and domain-specific vocabulary aligns with human technical writing, contrasting with AI’s tendency toward more compact or more heavily diversified long-word usage. Research reference: Mean word length: AI 5.86 vs Human 5.24.
— *“MobileNetV2 Accuracy VS Epoch Graph”*
- Discourse markers commonly found in human writing are present, such as "Thus" and "Therefore" usage in the text’s transitions, reflecting human-typical connective choices.
— *“Thus, they must be considered in sampling.”*





- Self-repetition rates for multi-gram sequences are higher, a pattern seen more in human writing, contributing to a familiar rhythm in the document (Human 3.5% vs AI 1.4% for 3-gram repetition).
— *“The user can upload an image of waste as input to the web-based waste classification system project.”*
- Pronoun subjects indicating personal voice: Several sentences employ pronoun subjects (e.g., "we obtained waste images" implied in 3.2 Data Collection, showing a direct, speaker-driven approach typical of human writing).
— *“For the study, we obtained waste images from open datasets such as Kaggle and TrashNet.”*
- Question marks occur more frequently in human writing, and the sample’s occasional interrogative phrasing aligns with that pattern (human-skewed behavior in punctuation usage).
— *“If this happens, it can have damaging impacts as the waste gets released ”*
- Concluding sections show a tightening of sentences, consistent with humans ending with concise statements rather than the uniform length often seen in AI outputs.
— *“[9] Ian Goodfellow, Yoshua Bengio, and Aaron Courville, Deep Learning, MIT Press, Cambridge, MA, USA, 2016.”*
- Adjacent sentence similarity is relatively higher, indicating that neighboring sentences share vocabulary and topics, a characteristic linked to human writing where ideas are revisited across nearby sentences.
— *“Also, It can be used for education and research purpose.”*
- Discourse markers demonstrating human style: The passage includes markers that align with human usage such as "The goal of this project is to develop" and other transitional phrases that resemble human discourse flow rather than rigid AI hedging.
— *“Specifically, the system identifies and categorizes waste.”*
- There is a tightening of conclusion-related language toward the end of sections, mirroring the human pattern of summarizing findings within each step rather than maintaining even pacing throughout (shorter or more definitive closing clauses near the end of sections).
— *“Training was performed for multiple epochs until the model achieved sta”*

Submission metadata

Submitted: 29 May 2026, 23:17	Timezone: UTC+05:30
Word count: 2326	Character count: 18023
Sentence count: 500	Language: English
Detector model version: HAI-DX v4.1	

These results indicate the probability of AI-generated content. They are not a definitive verdict.





How We Score

This report is generated by Humanize AI Detector. Scores reflect the likelihood that text is AI-generated and are not definitive. Detector accuracy: 96.3%, false-positive rate: 1.1% (benchmark: MAGE-2).

Mini-FAQ questions

• What does this % mean?

The percentage represents the model's confidence that the text was generated by an AI tool. Higher values indicate stronger AI signals; lower values indicate the text reads as human-written.

• Does this mean cheating?

No. This report is a probabilistic signal, not a definitive verdict. AI-assisted writing is widespread and legitimate in many contexts. Use this report alongside human judgment.

• Can I dispute a result?

Yes. False positives can occur on highly polished or formulaic human writing. Reach out to support@humanizeai.pro with the Report ID for a manual review.



● AI-generated

● AI-assisted / paraphrased

● Likely human

CHAPTER – 1

Classification of Recyclable Waste Using Deep Learning

INTRODUCTION

1.1 Overview

Waste management involves managing a variety of waste materials like gases, liquids or solids. These actions include transporting, gathering, knowing handling and recycling waste. Agriculture, industry, societies and other organization produce waste substances. Constantly changing waste properties in the process is significant. Thus, they must be considered in sampling. Measures to manage and control the production, storage, collection, transport, recovery and disposal of waste are taken using appropriate technology and scientific powers.

In order to stay sustainable, human societies must process an enormous amount of waste. Getting rid of waste is a huge issue as they keep producing waste regularly and some are tough to treat. Nuclear waste, for example, has specific treatment issues. In the same fashion, waste that is extremely poisonous also has its own treatment challenges. The volume and the characteristics both play a critical role in waste processing.

Recycling Waste Is Important.

The volume of waste substances is increasing at a very high rate due to major driving forces like urbanization, industrialization and increase in population. If this happens, it can have damaging impacts as the waste gets released in

huge volumes into the air, water and soil. Various sources produce waste that is harmful to the quality of life.

The annual statistic is more than 5.4 million tons of recyclable waste is sent sent to landfill or incinerated. Besides, the typical American produces about 4.4 pounds of garbage a day. Individuals from all walks of life generate recycling waste every day. So, the goal of the project “Web Based Waste Classification System” is to automate waste classification and stop recyclable waste going to a landfill.

Waste sorting at home, in the office and almost everywhere by different people is now common. Nonetheless, due to the various types and varieties of waste we generate, the classification of wastes is still a challenging task for the labourers. Therefore, developing a Classifier System can bring workers to a better solution for accurate waste classification.

The user can upload an image of waste as input to the web-based waste classification system project. After that, the system predicts the class of waste whether it is recyclable or bio-degradable using the model. The project employs top-notch image classification models which include CNN, VGG16, ResNet50, MobileNet and EfficientNet for extraction of features and classification.

We are designing a web-based system to classify wastes using deep learning. It assists with.

1.2 Problem Statement





A traditional method of disposal is the manual segregation of waste. However, it relies on human observation who may have trouble identifying it as many of its looks similar

1.2.1 Manual Classification Issues

- Needs the help of humans.Human errors are common
- Takes a long time.
- Managing waste on a large scale is not easy.

1.2.2 Similar Appearance of Waste Materials

Various waste materials show similarity:

- Color
- Shape
- Texture
- Outer Looks

This causes commotion in manual classification.

1.2.3 Lack of Automated Systems

After a thorough paper review conducted on the used solid waste, it was established that the geographic location, occupation and habit of the resident plays a key role in shaping solid waste.

1.2.4 Dataset Challenges

- Labeled datasets are limited
- Datasets are not balanced
- Image quality is different
- Light and background is different

Aim of the Project

The project aims to design a system that completely automates the segregating of waste using deep learning advancement. Specifically, the system identifies and categorizes waste.

1.3 Objectives

1. To design and implement automated waste classification system.
2. To apply online image preprocessing techniques.
3. Deep learning models will also extract richer features from this for further analysis.
4. To improve classification accuracy using transfer learning.
5. To compare multiple deep learning architectures.
6. To balance the dataset for better performance.
7. To reduce dependency on manual identification.
8. To spread awareness about proper waste segregation.
9. To develop a scalable and user-friendly system.

1.4 Applications

The proposed system has various applications in the real world:

- Smart Waste Management

Automatic segregation of waste must be done at homes, offices, schools, public places.

- Recycling Industry

The management of a scrap is a major issue in a manufacturing unit.

- Municipal Waste Management

These records aid local municipal authorities in enhancing waste collection and segregation systems in the region.

- Environmental Protection





Minimizes pollution and encourages proper waste disposal practices.

• Educational Purpose

Management and recycling can be taught in schools and colleges to make them aware of it.

• Smart City Applications

Smart waste management systems can be installed in smart cities.

• Public Awareness Systems

Provides disposal instructions and encourages people to follow proper waste segregation methods.

1.5 Scope of the Project

The goal of this project is to develop an intelligent system which is capable of detecting waste of different categories.

Future enhancements include:

- Real-time waste detection
- Multi-waste object detection
- Increasing the range of waste categories.
- Improved Dataset and Accuracy
- Environmental Monitoring and Analytics

Chapter-2

LITERATURE SURVEY

2.1 Introduction

The document discusses the waste classification system's research work done by various researchers. The waste classification system helps to sort various kinds of waste materials at the initial stage of waste management. Increasing interest for segregation of waste is being automated to enhance efficiency of waste classification for recycling. For waste segregation in the real world, scientists and engineers use machine learning and deep learning techniques. The technique of identifying object/scene activity in a digital image or video is called an image recognition system.

2.2 Research Paper 1

Title

Organic and Recyclable Waste Classification using Deep Learning Methods

Method Used

- VGG16
- VGG19
- ResNet18
- DenseNet121
- DenseNet169
- Simple CNN
- Transfer Learning

Working

- Images of waste are collected from Kaggle dataset.
- The image gets preprocessed and resized.
- Deep learning models are used to extract image features automatically.
- Models categorize waste as organic and recyclable.
- The best performing model is selected based on performance selection.

Accuracy

Using customized VGG16 with transfer learning 95.59%.

Advantages

- High classification accuracy.





- Performance is improved by transfer learning.
- Extraction of Features Automatically.
- Allows for several deep learning structures.

Limitations

- Needs a large training set.
- Accuracy affected by image quality.
- High computational requirements.

1.3 Research Paper 2 –

Title

Classification of Recyclable Waste using Deep Learning Approaches

Method Used

- ResNet-50
- Custom 5-Layer CNN

- Image Classification Techniques
- Deep Learning Frameworks

Accuracy

A custom CNN model achieves 91.84%

Advantages

- It is usable feature for web and mobile application integration.
- Reduced training data requirements
- Quicker training processes

Limitations

- Requires proper hyperparameter tuning
- Performance depends on dataset quality
- Deep models require more memory

TABLE 2.1 COMPARISION OF RESEARCH PAPER

Chapter-3

METHODOLOGY

3.1 System Workflow

Collecting images is the first step for the proposed waste classification system. The next step is pre-processing which further consists of two steps namely annotation and augmentation. After that the, train-test split will be done on the pre-processed data.

3.2 Data Collection

For the study, we obtained waste images from open datasets such as Kaggle and TrashNet. The dataset contains images of recyclable and biodegradable waste under different backgrounds and lighting conditions.

3.3 Image Preprocessing

By resizing the images, normalizing them, rotating and flipping them and applying augmentation, the model accuracy was improved and overfitting was reduced.

3.4 Model Selection

Deep learning models used for feature extraction and classification include CNN, VGG16, ResNet50, MobileNet and EfficientNet. Techniques of transfer learning were also included.

3.5 Model Training

The dataset was divided into training, validation, and testing datasets. Using Adam optimizer and categorical crossentropy loss function, the models were





trained with TensorFlow and Keras frameworks.

3.6 Model Evaluation

Metrics for the evaluation of the trained models were accuracy, validation loss, recall, and precision. We selected the top-performing model for implementation.

Chapter-4

SOFTWARE REQUIREMENTS

- Operating System: Windows 10/11 or Linux
- Programming Language: Python 3.14
- Development Platform: JupyterNotebook / VS Code / Google Colab
- Frameworks: TensorFlow.
- Frontend: HTML, CSS, JavaScript
- Backend: Flask
- Dataset Source: Kaggle and TrashNet datasets
- Hardware Requirement: Minimum 8 GB RAM, Intel i5 Processor or above

Chapter-5

IMPLEMENTATION DETAILS

IMPLEMENTATION DETAILS

5.1 Step 1: System Design Selection

The model proposed here is based on Deep Learning architectures for garbage classification. We created a simple convolutional neural network (CNN) model first due to the reason that these models are able to automatically learn important image features like shape, colour, texture, and even patterns of objects from waste images.

Following that, sophisticated architectures, such as VGG16, ResNet50, MobileNet, and EfficientNet, were examined on the input data to achieve enhanced classification accuracy and improved feature extraction capabilities. Due to better performance and higher accuracy all these versions of CNN are widely used in image classification tasks.

The proposed model will be a web-application under which a user may upload the image of the waste. Additionally, the trained model predicts if the waste is recyclable or biodegradable..

5.2 Step 2: Dataset Used

We gathered images of waste from publicly available data sets and online sources for this project. More specifically, we used kaggle and trashNet.

Images of different waste belonging to the following classes are available in the data.

- Waste that is recyclable
- (3 words): Biodegradable waste.

Images of waste were taken under various conditions ranging from the following –.

- Varied lighting conditions.
- People from diverse backgrounds.
- Various positions and perspectives.
- We have utilized images with various resolutions.

It makes the model stronger and this will ensure it works better in real life.

5.3 Step 3: Dataset Loading

The dataset applied to this project is downloaded from Kaggle. The dataset





consists of images of waste materials that can be used in the training and testing of deep learning models.

Source of dataset.

E-waste image dataset available on Kaggle

Python deep learning frameworks like Tensorflow and Keras load the dataset.

The image folders are organized and classified which makes the dataset suitable for supervised learning..

5.4 Step 4: Image Preprocessing

To improve the quality of the images, the preprocessing operations are applied before the model training. As a result, the model's ability to generalize has improved which reduces overfitting. Preprocessing Carry Out the Following Functions.

Change image size to 224×224 .

Scaling Pixels.

Soundproofing Solution

Data Enhancement.

Turning and Rotating.

Zoom Change.

5.5 Step 5: Image Reshaping

All images are resized to a fixed dimension for compatibility with deep learning models. The input images are converted to tensor format of dimension size compatible for CNN and Vision Transformer models training and testing.

5.6 Step 6: Label Encoding

Each waste category is assigned a numerical label.

For example:

- Recyclable 0
- Biodegradable 1

The labels are converted into one-hot encoded vectors so that the deep learning models can correctly classify multiple categories during supervised learning.

5.7 Step 7: Data Splitting

The learning model parameters are defined using the training dataset and the evaluation of the performance of learned model on unseen Waste images is done by testing dataset.

5.8 Step 8: Model Training

Various Deep learning models such as:

- CNN
- VGG16
- ResNet50
- MobileNet
- EfficientNet

were trained using the prepared dataset.

During training:

- Forward propagation computes predictions
- Backpropagation updates weights
- Adam optimizer minimizes loss
- Categorical Crossentropy is used as the loss function

Transfer learning with pre-trained ImageNet weights was also applied to improve feature extraction and reduce training time.

Training was performed for multiple epochs until the model achieved stable





accuracy and reduced loss.

5.9 Step 9: Dataset Balancing and Optimization

Initially, the dataset was imbalanced, which caused poor accuracy and biased predictions.

To solve this issue:

- Data augmentation techniques were applied
- Smaller classes were increased
- Duplicate and unnecessary images were removed
- Class balancing techniques were used

After balancing the dataset:

- Model performance improved
- Overfitting was reduced
- Classification became more accurate

5.10 Step 10: Model Testing

Various performance metrics were used to evaluate the trained models:

- Accuracy
- Loss
- Validation Accuracy
- Training Accuracy
- Precision
- Recall

It was observed that there was evaluation comparative of performance of different model. Among all the models, transfer learning models like MobileNet outperformed basic CNN models in terms of a score. The selected model is able to achieve a good classification accuracy and performance is achieved.

5.11 Step 11: Output Generation

The final trained system accepts a waste image as input and predicts the most probable waste category as output.

The system automatically performs:

- Image preprocessing
- Feature extraction
- Classification

before displaying the result.

The output includes:

- Predicted waste category

This helps users understand how to dispose of waste correctly and promotes better waste management practices.

Chapter-6

RESULTS

Figure 6.1: Web interface showing recyclable waste prediction with confidence score and disposal guidance.

Figure 6.2: Web interface displaying non-recyclable waste classification with disposal instructions.

Figure 6.3: Confusion matrix showing the accuracy of the waste classification model.

Figure 6.4: MobileNetV2 Accuracy VS Epoch Graph

CONCLUSION





The project entitled “Web Based Waste Classification System Using Deep Learning” shows the use of Artificial Intelligence and Deep Learning in the Waste Management. The system automatically classifies waste images as recyclable or biodegradable. The project employs state-of-the-art image classification techniques and deep learning architectures. The initial phase of testing shall involve the use of multiple deep learning models CNN, VGG16, ResNet50, MobileNet, EfficientNet to classify waste. If we develop a dataset which is free from above issues, it could be more effective at solving waste recognition and classification problem.

In order to decrease the issues with these data, a few preprocessing techniques are applied followed by data augmentation. To enhance overall classification accuracy, further balancing of the dataset was performed. The project also focuses on preparation of an appropriate dataset, image preprocessing, data augmentation, transfer learning, balancing the dataset to reduce training-testing bias. It was also shown how models could be optimized.

The waste classification is done with very less human effort. It will help in building better smart waste management solutions. Users will also get live hands-on experience of deep learning, computer vision, transfer learning and image classification techniques.

The system has significant potential for implementation in the real-world. It can be employed in the intelligent waste management framework, recycling sector, municipal waste management, smart city missions, environmental awareness initiatives and more. Also, It can be used for education and research purpose.

Also, the system being web-based, the users can easily use it from different devices without the hassle of complex installation.

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